## Supplementary material

### S1: Search strategy

((((“cognitive training” or “brain training” or “attention training” or “reasoning training” or “memory training” or “mental training” or “mental skills training” or “neurocognitive training” OR “executive function training” OR “attentional control training”) OR (“cognitive exercise” or “brain exercise” or “memory exercise” or “attention exercise” or “reasoning exercise”) OR (“cognitive stimulation” or “memory stimulation” or “memory enhance$” or “cognitive enhanc$” OR “executive function enhancement”) OR (“cognitive activit$” or “mental activit$”)
OR (speed and processing and training) OR mnemonic$ OR (“video game$” or videogame$ or wii or “computer game$” or “virtual reality”) OR (“cognitive intervention$” or “neurocognitive intervention$”)) AND ((exercis$ OR sport$ OR “physical fitness”) OR (“aerobic exercis$” or “aerobic train$” or “aerobic fitness” or “aerobic program$”) OR (“resistance exercis$” or “resistance train$” or “anaerobic exercis$” or “anaerobic train$” or “resistance program$”) OR (physical or aerobic or endurance or cardiorespiratory or cardiovascular or resistance or strength) OR (bicycl$ or “bike rid$” or “bicycle rid$”)))

OR

(((multimodal or multidomain or multicomponent or “multi-modal” or “multi-domain” or “multi-component” or “dual task” or “dual-task” or “tai chi” or danc$) OR (exergame$ or “active video game$” or “active videogame$” or kinect or “active play” or “interactive video”)) AND (cognitive adj2 physical) AND (cognition or cognitive or memory or executive function$ or “executive control” or attention or visuospatial or “processing speed” or language)))

AND (“older adults” or elder$ or senior$ or adult$ or older or ag?ing)

| **Table S2. Assessment tools used in the included studies and their classification into cognitive (executive functions, memory, speed, attention, global cognition, and composite scores), and physical functions (fitness, balance, and strength).** | | | |
|---|---|---|---|
| **Authors** | **Year** | **Test** | **Function** |
| Adcock et al | 2020 | 2min stepping test | fitness |
| Adcock et al | 2020 | 30 s chair rises test | strength |
| Adcock et al | 2020 | Cycle duration CV [%] | balance |
| Adcock et al | 2020 | Digit span backward score | executive |
| Adcock et al | 2020 | Digit span forward | memory |
| Adcock et al | 2020 | Digit span forward score | memory |
| Adcock et al | 2020 | Extended balance test of SPPB | balance |
| Adcock et al | 2020 | Gait Speed mean [m/s] | fitness |
| Adcock et al | 2020 | Stride length CV [%] | balance |
| Adcock et al | 2020 | Stride length mean [m] | balance |
| Adcock et al | 2020 | TMT A -s | speed |
| Adcock et al | 2020 | TMT B-s | executive |
| Adcock et al | 2020 | Toe clearance CV [%] | balance |
| Adcock et al | 2020 | Toe clearance mean [cm] | balance |
| Adcock et al | 2020 | Victoria 3-2 | executive |
| Adcock et al | 2020 | Victoria Stroop 1 - time | speed |
| Adcock et al | 2020 | Victoria Stroop 2 - time | speed |
| Adcock et al | 2020 | Victoria Stroop 3 - time | executive |
| Anderson-Hanley et al | 2012 | Clock drawing | global |
| Anderson-Hanley et al | 2012 | Color trail test | executive |
| Anderson-Hanley et al | 2012 | COWAT-categories | language |
| Anderson-Hanley et al | 2012 | COWAT-total | language |
| Anderson-Hanley et al | 2012 | Digit Span backward | executive |
| Anderson-Hanley et al | 2012 | Figure copy | global |
| Anderson-Hanley et al | 2012 | Figure copy delayed | global |
| Anderson-Hanley et al | 2012 | Letter Digit Symbol Test | attention |
| Anderson-Hanley et al | 2012 | RAVLT delayed | memory |
| Anderson-Hanley et al | 2012 | RAVLT immediate | memory |
| Anderson-Hanley et al | 2012 | RAVLT_5trials | memory |
| Anderson-Hanley et al | 2012 | Stroop | executive |
| Andrieu et al | 2017 | Category Naming Test | language |
| Andrieu et al | 2017 | Composite z score | composite |
| Andrieu et al | 2017 | COWAT | language |
| Andrieu et al | 2017 | DSST | speed |
| Andrieu et al | 2017 | Free and Cued Selective Reminding | memory |
| Andrieu et al | 2017 | Gait speed | fitness |
| Andrieu et al | 2017 | MMSE | global |
| Andrieu et al | 2017 | MMSE orientation | global |
| Andrieu et al | 2017 | SPPB | fitness |
| Andrieu et al | 2017 | TMT-A | speed |
| Andrieu et al | 2017 | TMT-B | executive |
| Andrieu et al | 2017 | visual analogue scale | memory |
| Bamidis et al | 2015 | Composite z score | composite |
| Barban et al | 2017 | RAVLT delayed | memory |

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|---|---|---|---|
| Desjardins-Crepeau et al | 2016 | 6-min walk | fitness |
| Desjardins-Crepeau et al | 2016 | Baddeley dual task - single | speed |
| Desjardins-Crepeau et al | 2016 | Baddeley dual-task interfern index | executive |
| Desjardins-Crepeau et al | 2016 | Chair stand test | strength |
| Desjardins-Crepeau et al | 2016 | Color-word inhibition | executive |
| Desjardins-Crepeau et al | 2016 | Color-word interference_color | speed |
| Desjardins-Crepeau et al | 2016 | Color-word interference_reading | speed |
| Desjardins-Crepeau et al | 2016 | Color-word task switching | executive |
| Desjardins-Crepeau et al | 2016 | Handgrip strength | strength |
| Desjardins-Crepeau et al | 2016 | Modified phys performance test | fitness |
| Desjardins-Crepeau et al | 2016 | RAVLT delayed | memory |
| Desjardins-Crepeau et al | 2016 | RAVLT immediate | memory |
| Desjardins-Crepeau et al | 2016 | RAVLT total | memory |
| Desjardins-Crepeau et al | 2016 | TMT A | speed |
| Desjardins-Crepeau et al | 2016 | TMT B | executive |
| Desjardins-Crepeau et al | 2016 | TUG | balance |
| Eggenberger et al | 2015 | Age concentration A | attention |
| Eggenberger et al | 2015 | Age concentration B | attention |
| Eggenberger et al | 2015 | Digit Span forward | memory |
| Eggenberger et al | 2015 | DSST | speed |
| Eggenberger et al | 2015 | Executive control task | executive |
| Eggenberger et al | 2015 | PALT | memory |
| Eggenberger et al | 2015 | Story recall | memory |
| Eggenberger et al | 2015 | TMT A | speed |
| Eggenberger et al | 2015 | TMT B | executive |
| Fabre et al | 2002 | Digit Span forward | memory |
| Fabre et al | 2002 | Logical memory - immediate | memory |
| Fabre et al | 2002 | Logical memory information | memory |
| Fabre et al | 2002 | Logical memory mental control | executive |
| Fabre et al | 2002 | Logical memory orientation | memory |
| Fabre et al | 2002 | Logical memory visual reprod | memory |
| Fabre et al | 2002 | O2 pulse | fitness |
| Fabre et al | 2002 | O2pulse max | fitness |
| Fabre et al | 2002 | PALT | memory |
| Fabre et al | 2002 | VO2 | fitness |
| Fabre et al | 2002 | VOX2max | fitness |
| Fabre et al | 2002 | Wais memory quotient | memory |
| Gill et al | 2016 | Auditory verbal learning-learning | memory |
| Gill et al | 2016 | Autitory verbal learning-recall | memory |
| Gill et al | 2016 | DSST | speed |
| Gill et al | 2016 | Verbal fluency category | language |
| Gill et al | 2016 | Verbal fluency letter | language |
| Gschwind et al | 2015 | 10-m walk-single task | fitness |
| Gschwind et al | 2015 | Attention network test-alert | attention |
| Gschwind et al | 2015 | Attention network test-conflict | attention |
| Gschwind et al | 2015 | Attention network test-orient | attention |
| Gschwind et al | 2015 | Attention network test-RT | attention |

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|---|---|---|---|
| Gschwind et al | 2015 | Coordinated stability | balance |
| Gschwind et al | 2015 | Digit Span backward | executive |
| Gschwind et al | 2015 | DSST | speed |
| Gschwind et al | 2015 | Handgrip strength | strength |
| Gschwind et al | 2015 | Knee extension | strength |
| Gschwind et al | 2015 | Maximum balance range-antero posterior | balance |
| Gschwind et al | 2015 | Melbourne edge test | balance |
| Gschwind et al | 2015 | Propioception | balance |
| Gschwind et al | 2015 | Sensor-based chair stand test | strength |
| Gschwind et al | 2015 | Sensor-based full tandem stance | balance |
| Gschwind et al | 2015 | Sensor-based near tandem stance | balance |
| Gschwind et al | 2015 | Sensor-based semi tandem stance | balance |
| Gschwind et al | 2015 | SPPB | fitness |
| Gschwind et al | 2015 | Sway-area | balance |
| Gschwind et al | 2015 | TMT A | speed |
| Gschwind et al | 2015 | TMT B | executive |
| Gschwind et al | 2015 | TUG | balance |
| Gschwind et al | 2015 | Victoria Stroop-efficacy score | executive |
| Gschwind et al | 2015 | Victoria Stroop-intrusions | executive |
| Hiyamizu et al | 2012 | Chair stand test | strength |
| Hiyamizu et al | 2012 | Functional reach test | balance |
| Hiyamizu et al | 2012 | Stroop ACC | executive |
| Hiyamizu et al | 2012 | Sway - eyes closed | balance |
| Hiyamizu et al | 2012 | Sway - eyes open | balance |
| Hiyamizu et al | 2012 | TMT A | speed |
| Hiyamizu et al | 2012 | TMT B | executive |
| Hiyamizu et al | 2012 | TMT B-A | executive |
| Hiyamizu et al | 2012 | TUG | balance |
| Htut et al | 2018 | Five times sit to stand | strength |
| Htut et al | 2018 | Handgrip left | strength |
| Htut et al | 2018 | Handgrip right | strength |
| Htut et al | 2018 | MoCA | global |
| Htut et al | 2018 | TUG | balance |
| Jardim et al | 2021 | 30-s chair stand | strength |
| Jardim et al | 2021 | 6-m walk | fitness |
| Jardim et al | 2021 | CERARD word list - evocation | memory |
| Jardim et al | 2021 | CERARD word list - inm | memory |
| Jardim et al | 2021 | CERARD word list - recognition | memory |
| Jardim et al | 2021 | PALT - nº of patterns | memory |
| Jardim et al | 2021 | PALT - stages completed | memory |
| Jardim et al | 2021 | PALT - total | memory |
| Jardim et al | 2021 | Rapid visual processing | attention |
| Jardim et al | 2021 | TUG | balance |
| Jardim et al | 2021 | Walking m/s | fitness |
| Jehu et al | 2017 | Counting backward (TUGcog) | executive |
| Jehu et al | 2017 | TUG | balance |
| Joubert & Chainay | 2019 | Complex Span task - ACC | executive |

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|---|---|---|---|
| Joubert & Chainay | 2019 | Complex Span task - RT | executive |
| Joubert & Chainay | 2019 | Flanker task - ACC | executive |
| Joubert & Chainay | 2019 | Flanker task - RT | executive |
| Joubert & Chainay | 2019 | Plus Minus task - ACC | executive |
| Joubert & Chainay | 2019 | Plus Minus task - RT | executive |
| Joubert & Chainay | 2019 | RAVLT- lexical | memory |
| Joubert & Chainay | 2019 | RAVLT-categories | memory |
| Joubert & Chainay | 2019 | TMT B-A | executive |
| Joubert & Chainay | 2019 | Updated Span task - ACC | executive |
| Joubert & Chainay | 2019 | Updated Span task - RT | executive |
| Kitazawa et al | 2015 | Dementia Assessment Scale | global |
| Kitazawa et al | 2015 | Touch-M - visuospatial | memory |
| Kitazawa et al | 2015 | TUG | balance |
| Laatar et al | 2018 | 30-s chair stand test | strength |
| Laatar et al | 2018 | CoP x | balance |
| Laatar et al | 2018 | CoP y | balance |
| Laatar et al | 2018 | Functional reach test | balance |
| Laatar et al | 2018 | Gait speed | fitness |
| Laatar et al | 2018 | Simple reaction time | speed |
| Laatar et al | 2018 | TUG | balance |
| Legault et al | 2011 | 1-back | memory |
| Legault et al | 2011 | 2-back | executive |
| Legault et al | 2011 | Flanker task | executive |
| Legault et al | 2011 | HVLT delayed | memory |
| Legault et al | 2011 | HVLT immediate | memory |
| Legault et al | 2011 | HVLT suppl score | memory |
| Legault et al | 2011 | HVTL total | memory |
| Legault et al | 2011 | Self-ordered pointing task | executive |
| Legault et al | 2011 | Task switching | executive |
| Legault et al | 2011 | TMT B-A | executive |
| Linde & Alfermann | 2014 | d2 test of attention | attention |
| Linde & Alfermann | 2014 | DSST | speed |
| Linde & Alfermann | 2014 | Leistungs-Pruf-System 50+ Reasoning | speed |
| Linde & Alfermann | 2014 | Leistungs-Pruf-System 50+ Spatial relatio | speed |
| Linde & Alfermann | 2014 | TMT A | speed |
| Linde & Alfermann | 2014 | V02max | fitness |
| Linde & Alfermann | 2014 | Word list test | memory |
| Maillot & Hartley | 2012 | 6-Min Walk test-distance | fitness |
| Maillot & Hartley | 2012 | 6-Min Walk test-max HR | fitness |
| Maillot & Hartley | 2012 | 6-Min Walk test-mean HR | fitness |
| Maillot & Hartley | 2012 | 8-Foot Up-and Go test | fitness |
| Maillot & Hartley | 2012 | Arm curls | strength |
| Maillot & Hartley | 2012 | Back Scratch test – lower left | fitness |
| Maillot & Hartley | 2012 | Back Scratch test – lower right | fitness |
| Maillot & Hartley | 2012 | Back Scratch test – upper left | fitness |
| Maillot & Hartley | 2012 | Back Scratch test – upper right | fitness |
| Maillot & Hartley | 2012 | Cancellation test | speed |

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|---|---|---|---|
| Maillot & Hartley | 2012 | Chair stand test | strength |
| Maillot & Hartley | 2012 | Directional Headings test | executive |
| Maillot & Hartley | 2012 | DSST | speed |
| Maillot & Hartley | 2012 | Letter Sets test | executive |
| Maillot & Hartley | 2012 | Matrix reasoning test | executive |
| Maillot & Hartley | 2012 | Mental rotation test | executive |
| Maillot & Hartley | 2012 | Number comparison test | speed |
| Maillot & Hartley | 2012 | Reaction time test – choice | speed |
| Maillot & Hartley | 2012 | Reaction time test – simple | speed |
| Maillot & Hartley | 2012 | Spatial Span test | executive |
| Maillot & Hartley | 2012 | Spatial Span test - backward | executive |
| Maillot & Hartley | 2012 | Stroop incongruent | executive |
| Maillot & Hartley | 2012 | Stroop switching | executive |
| Maillot & Hartley | 2012 | TMT B-A | executive |
| Marmeleira et al. | 2009 | Dual-task movement time | attention |
| Marmeleira et al. | 2009 | Dual-task reaction time | attention |
| Marmeleira et al. | 2009 | Dual-task response time | attention |
| Marmeleira et al. | 2009 | Foot tap test (Lower limb mobility) | fitness |
| Marmeleira et al. | 2009 | Functional reach test | balance |
| Marmeleira et al. | 2009 | Self-only in motion. Absolute errors | attention |
| Marmeleira et al. | 2009 | Self-only in motion. Constant errors | attention |
| Marmeleira et al. | 2009 | Self-only in motion. Variable errors | attention |
| Marmeleira et al. | 2009 | Single-task. Movement time | speed |
| Marmeleira et al. | 2009 | Single-task. Reaction time | speed |
| Marmeleira et al. | 2009 | Single-task. Response time | speed |
| Marmeleira et al. | 2009 | Stroop - incongruent | executive |
| Marmeleira et al. | 2009 | Stroop - interference | executive |
| Marmeleira et al. | 2009 | Target-only in motion. Absolute errors | attention |
| Marmeleira et al. | 2009 | Target-only in motion. Constant errors | attention |
| Marmeleira et al. | 2009 | Target-only in motion. Variable errors | attention |
| Marmeleira et al. | 2009 | Three-choice reaction time | speed |
| Marmeleira et al. | 2009 | TMT B errors | executive |
| Marmeleira et al. | 2009 | TMT B s | executive |
| Marmeleira et al. | 2009 | TUG | balance |
| Marmeleira et al. | 2009 | Two-choice reaction time | speed |
| Marmeleira et al. | 2009 | Useful Field of View - divided att | attention |
| Marmeleira et al. | 2009 | Useful Field of View - selective att | attention |
| Marmeleira et al. | 2009 | Useful Field of View - speed | speed |
| McDaniel et al | 2014 | Cooking Breakfast Task- Ideal Performance | memory |
| McDaniel et al | 2014 | Cooking Breakfast Task- Number of Table S | memory |
| McDaniel et al | 2014 | Cooking Breakfast Task- Stopping Time Ran | memory |
| McDaniel et al | 2014 | Memory for Health Information Task- Corre | memory |
| McDaniel et al | 2014 | Memory for Health Information Task- FAs t | memory |
| McDaniel et al | 2014 | Memory for Health Information Task- Sourc | memory |
| McDaniel et al | 2014 | Virtual Week Task - irregular | attention |
| McDaniel et al | 2014 | Virtual Week Task - regular | attention |
| McDaniel et al | 2014 | Virtual Week Task – time based | attention |

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|---|---|---|---|
| McDaniel et al | 2014 | VO2peak | fitness |
| Morita et al | 2018 | Maximal step length | fitness |
| Morita et al | 2018 | Modified Mini-Mental State (3MS) | global |
| Morita et al | 2018 | Quad. Muscle strength | strength |
| Morita et al | 2018 | Single-leg standing | balance |
| Morita et al | 2018 | TUG | balance |
| Ng et al | 2017 | RBANS - attention | attention |
| Ng et al | 2017 | RBANS - language | language |
| Ng et al | 2017 | RBANS - total | composite |
| Ng et al | 2017 | RBANS - visuospatial | executive |
| Ng et al | 2017 | RBANS – delayed | memory |
| Ng et al | 2017 | RBANS – immediate | memory |
| Ngandu et al | 2015 | Neuropsychological test battery - Executive Functions | executive |
| Ngandu et al | 2015 | Neuropsychological test battery - Memory | memory |
| Ngandu et al | 2015 | Neuropsychological test battery - Processing speed | speed |
| Ngandu et al | 2015 | Neuropsychological test battery – Memory short version | memory |
| Ngandu et al | 2015 | Neuropsychological test battery (NTB) – total | composite |
| Nilsson et al | 2020 | Episodic memory spatial + verbal | memory |
| Nilsson et al | 2020 | ETS kit verbal inference + BIS analogies + Syllogisms | language |
| Nilsson et al | 2020 | n-back + Running span trained | speed |
| Nilsson et al | 2020 | n-back + Running span untrained | speed |
| Nilsson et al | 2020 | Numerical and spatial updating | speed |
| Nilsson et al | 2020 | Perceptual matching 1+ 2 | speed |
| Nilsson et al | 2020 | Raven's Progressive Matrices | executive |
| Nilsson et al | 2020 | Task switching 1 + 2 | speed |
| Nishiguchi et al | 2015 | 0-back - face-ACC | speed |
| Nishiguchi et al | 2015 | 0-back - face-ms | speed |
| Nishiguchi et al | 2015 | 0-back - face+location ACC | speed |
| Nishiguchi et al | 2015 | 0-back - face+location-ms | speed |
| Nishiguchi et al | 2015 | 0-back - location-ACC | speed |
| Nishiguchi et al | 2015 | 0-back - location-ms | speed |
| Nishiguchi et al | 2015 | 1-back face+location-ms | executive |
| Nishiguchi et al | 2015 | 1-back - face-ms | executive |
| Nishiguchi et al | 2015 | 1-back - face+location-ms | executive |
| Nishiguchi et al | 2015 | 1-back - location-ACC | executive |
| Nishiguchi et al | 2015 | 1-back - location-ms | executive |
| Nishiguchi et al | 2015 | 10-m walk test | fitness |
| Nishiguchi et al | 2015 | Chair stand test | strength |
| Nishiguchi et al | 2015 | Daily steps | fitness |
| Nishiguchi et al | 2015 | Logical memory - delayed | memory |
| Nishiguchi et al | 2015 | Logical memory - immediate | memory |
| Nishiguchi et al | 2015 | MMSE | global |
| Nishiguchi et al | 2015 | TMT B-A | executive |
| Nishiguchi et al | 2015 | TUG | balance |
| Nocera et al | 2020 | Digit Span backward | executive |
| Nocera et al | 2020 | Digit Span forward | memory |

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|---|---|---|---|
| Nocera et al | 2020 | Letter fluency | language |
| Nocera et al | 2020 | n-back-ACC | executive |
| Nocera et al | 2020 | n-back-ms | executive |
| Nocera et al | 2020 | Semantic fluency | language |
| Nocera et al | 2020 | Single gait | fitness |
| Nocera et al | 2020 | SPPB | fitness |
| Nocera et al | 2020 | Stroop | executive |
| Nocera et al | 2020 | TMT A | speed |
| Nocera et al | 2020 | TMT B | executive |
| Nocera et al | 2020 | VO2 | fitness |
| Norouzi et al | 2019 | Berg Balance Scale | balance |
| Norouzi et al | 2019 | n-back | executive |
| Oswald et al | 2006 | Composite z score - cognitive | composite |
| Phirom et al | 2020 | MoCA | global |
| Phirom et al | 2020 | Physiological Profile Assessment – knee extension strength | strength |
| Phirom et al | 2020 | Physiological Profile Assessment – Sway | balance |
| Phirom et al | 2020 | TUG – single task | balance |
| Pieramico et al | 2012 | Babcock Story – Delayed Recall | memory |
| Pieramico et al | 2012 | Babcock Story – Immediate Recall | memory |
| Pieramico et al | 2012 | Babcock Story Recall Test | memory |
| Pieramico et al | 2012 | Frontal Assessment Battery | global |
| Pieramico et al | 2012 | MMSE | global |
| Pieramico et al | 2012 | Phonological Fluency test | language |
| Pieramico et al | 2012 | TMT A | speed |
| Pieramico et al | 2012 | TMT B | executive |
| Pieramico et al | 2012 | TMT B-A | executive |
| Rahe et al (a) | 2015 | 30 s Chair stand | strength |
| Rahe et al (a) | 2015 | 6 minute walk test/2Min step test | fitness |
| Rahe et al (a) | 2015 | 8 foot up and go test | fitness |
| Rahe et al (a) | 2015 | Arm curl | strength |
| Rahe et al (a) | 2015 | Brief Test of Attention | attention |
| Rahe et al (a) | 2015 | Chair sit and reach test | fitness |
| Rahe et al (a) | 2015 | Complex Figure Test- memory | memory |
| Rahe et al (a) | 2015 | DemTect - composite | composite |
| Rahe et al (a) | 2015 | DemTect subtest supermarket/animal | language |
| Rahe et al (a) | 2015 | DemTect-delayed recall | memory |
| Rahe et al (a) | 2015 | DemTect-immediate recall | memory |
| Rahe et al (a) | 2015 | Digit Span backward | executive |
| Rahe et al (a) | 2015 | Overal fitness | fitness |
| Rahe et al (a) | 2015 | Regensburger Wort Flüssigkeits-Test – Fluidez verbal | language |
| Rahe et al (a) | 2015 | Stroop | executive |
| Rahe et al (b) | 2015 | Brief Test of Attention | attention |
| Rahe et al (b) | 2015 | Complex Figure Test- memory | global |
| Rahe et al (b) | 2015 | COWAT | language |
| Rahe et al (b) | 2015 | TMT B/A | executive |
| Raichlen et al | 2020 | serially subtract 7's beginning at 500 | executive |

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|---|---|---|---|
| Raichlen et al | 2020 | Stride duration | balance |
| Raichlen et al | 2020 | Stride duration variability | balance |
| Raichlen et al | 2020 | Stride length | balance |
| Raichlen et al | 2020 | Stride length variability | balance |
| Raichlen et al | 2020 | Stride velocity | balance |
| Raichlen et al | 2020 | Stride velocity variability | balance |
| Romera-Liebana et al | 2018 | Abstraction of word pairs | language |
| Romera-Liebana et al | 2018 | Animal naming test | language |
| Romera-Liebana et al | 2018 | Designation of images | language |
| Romera-Liebana et al | 2018 | Designation of names | language |
| Romera-Liebana et al | 2018 | Evocation of words | language |
| Romera-Liebana et al | 2018 | Functional reach test | fitness |
| Romera-Liebana et al | 2018 | Handgrip | strength |
| Romera-Liebana et al | 2018 | SPPB | fitness |
| Romera-Liebana et al | 2018 | Unipodal station | fitness |
| Romera-Liebana et al | 2018 | Verbal memory delayed | memory |
| Romera-Liebana et al | 2018 | Verbal memory immediate | memory |
| Salazar et al | 2014 | Cadence | balance |
| Salazar et al | 2014 | Double support | balance |
| Salazar et al | 2014 | Step width | balance |
| Salazar et al | 2014 | Stride length | balance |
| Salazar et al | 2014 | Substracting digits backwards | executive |
| Salazar et al | 2014 | Swing | balance |
| Salazar et al | 2014 | Walk speed | fitness |
| Schoene et al (a) | 2013 | Alternate Step test | balance |
| Schoene et al (a) | 2013 | Chair stand test | strength |
| Schoene et al (a) | 2013 | Choice stepping reaction time. Movement time. | speed |
| Schoene et al (a) | 2013 | Choice stepping reaction time. Reaction time. | speed |
| Schoene et al (a) | 2013 | Choice stepping reaction time. Total response time. | speed |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). anteroposterior | balance |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). central | balance |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). medio-lateral | balance |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). Global | balance |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). Lower extremity strength | strength |
| Schoene et al (a) | 2013 | Physiological Profile Assessment (PPA). Propioception of lower extremities | balance |
| Schoene et al (a) | 2013 | Step inhibition test - s | executive |
| Schoene et al (a) | 2013 | Step inhibition test errors | executive |
| Schoene et al (a) | 2013 | Step inhibition test time/trials | executive |
| Schoene et al (a) | 2013 | TMT A | speed |
| Schoene et al (a) | 2013 | TUG | balance |
| Schoene et al (b) | 2015 | Attentional network test - alert | attention |
| Schoene et al (b) | 2015 | Attentional network test - executive | executive |
| Schoene et al (b) | 2015 | Attentional network test - orientation | attention |
| Schoene et al (b) | 2015 | Choice stepping movement time test | speed |
| Schoene et al (b) | 2015 | Choice stepping reaction time test | speed |

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|---|---|---|---|
| Schoene et al (b) | 2015 | Digit letter maximum | speed |
| Schoene et al (b) | 2015 | Digit letter mean | speed |
| Schoene et al (b) | 2015 | Digit letter minimum | speed |
| Schoene et al (b) | 2015 | Digit Span backward | executive |
| Schoene et al (b) | 2015 | Hand reaction time test | speed |
| Schoene et al (b) | 2015 | Mental rotation - errors | executive |
| Schoene et al (b) | 2015 | Mental rotation - TR | executive |
| Schoene et al (b) | 2015 | Stroop - errors | executive |
| Schoene et al (b) | 2015 | Stroop - TR | executive |
| Schoene et al (b) | 2015 | Stroop Stepping Test - TR | executive |
| Schoene et al (b) | 2015 | Stroop Stepping Test -errors | executive |
| Schoene et al (b) | 2015 | TMT A | speed |
| Schoene et al (b) | 2015 | TMT B | executive |
| Schoene et al (b) | 2015 | TMT B/A | executive |
| Shah et al | 2014 | 1-back | memory |
| Shah et al | 2014 | Borg's scale | balance |
| Shah et al | 2014 | COWAT | language |
| Shah et al | 2014 | Detection (DET) | speed |
| Shah et al | 2014 | Groton Maze learning | memory |
| Shah et al | 2014 | Immediate Recall | memory |
| Shah et al | 2014 | Incremental Shuttle Walk test | fitness |
| Shah et al | 2014 | long term delayed recall | memory |
| Shah et al | 2014 | short term delayed recall | memory |
| Shah et al | 2014 | Sum of Strength (kgs.) | strength |
| Shah et al | 2014 | Visual Memory - index score | memory |
| Shatil et al | 2013 | CogniFit avoiding distractors | executive |
| Shatil et al | 2013 | CogniFit divided attention | attention |
| Shatil et al | 2013 | CogniFit global visual memory | memory |
| Shatil et al | 2013 | CogniFit inhibition | executive |
| Shatil et al | 2013 | CogniFit naming | language |
| Shatil et al | 2013 | CogniFit planning | executive |
| Shatil et al | 2013 | CogniFit processing speed | speed |
| Shatil et al | 2013 | CogniFit self-awareness | executive |
| Shatil et al | 2013 | CogniFit shifting | executive |
| Shatil et al | 2013 | CogniFit time estimation | executive |
| Shatil et al | 2013 | CogniFit visual scanning | attention |
| Shatil et al | 2013 | CogniFit working memory | executive |
| Takeuchi et al | 2020 | 0-back | memory |
| Takeuchi et al | 2020 | 2-back | executive |
| Takeuchi et al | 2020 | Digit cancellation task | attention |
| Takeuchi et al | 2020 | Digit span | executive |
| Takeuchi et al | 2020 | Frontal lobe and executive function | executive |
| Takeuchi et al | 2020 | Logical memory | memory |
| Takeuchi et al | 2020 | Raven's Progressive Matrices | executive |
| Takeuchi et al | 2020 | Semantic fluency | language |
| Takeuchi et al | 2020 | Symbol search | speed |
| Teixeira et al | 2013 | Digit Span backward | executive |

| | | | |
|---|---|---|---|
| Teixeira et al | 2013 | Digit Span forward | memory |
| Teixeira et al | 2013 | MMSE | global |
| Teixeira et al | 2013 | Modified Card Sorting Test - errors | executive |
| Teixeira et al | 2013 | Modified Card Sorting Test - errors adjusted for age | executive |
| Teixeira et al | 2013 | Toulouse-Pierón Concentrated Attention Test - hits | attention |
| Teixeira et al | 2013 | Toulouse-Pierón Concentrated Attention Test - TR | attention |
| Theill et al. | 2013 | Continuous Performance task | attention |
| Theill et al. | 2013 | DSST | speed |
| Theill et al. | 2013 | Dual task (WM+gait) - errors | executive |
| Theill et al. | 2013 | Dual task (WM+gait) - hits | executive |
| Theill et al. | 2013 | Executive control task | executive |
| Theill et al. | 2013 | Gait variability – single task | fitness |
| Theill et al. | 2013 | Gait velocity – single task | fitness |
| Theill et al. | 2013 | Operation span test | memory |
| Theill et al. | 2013 | PALT | memory |
| Theill et al. | 2013 | Raven's Progressive Matrices | executive |
| Van het Reve & de Bruin | 2014 | TMT A | speed |
| Van het Reve & de Bruin | 2014 | TMT B | executive |
| Van het Reve & de Bruin | 2014 | Vienna Test System – divided attention, lower channel | attention |
| Van het Reve & de Bruin | 2014 | Vienna Test System – divided attention, upper channel | attention |
| Wollesen et al (a) | 2017 | Gait-line left | balance |
| Wollesen et al (a) | 2017 | Gait-line right | balance |
| Wollesen et al (a) | 2017 | SPPB | balance |
| Wollesen et al (a) | 2017 | Step length left | balance |
| Wollesen et al (a) | 2017 | Step length right | balance |
| Wollesen et al (a) | 2017 | Step width | balance |
| Wollesen et al (a) | 2017 | Stroop | executive |
| Wollesen et al (a) | 2017 | Stroop dual-task (during walking) | executive |
| Wollesen et al (b) | 2017 | Gait-line left | balance |
| Wollesen et al (b) | 2017 | Gait-line right | balance |
| Wollesen et al (b) | 2017 | SPPB | balance |
| Wollesen et al (b) | 2017 | Step length left | balance |
| Wollesen et al (b) | 2017 | Step length right | balance |
| Wollesen et al (b) | 2017 | Step width | balance |
| Wollesen et al (b) | 2017 | Stroop | executive |
| Wollesen et al (b) | 2017 | Stroop dual-task (during walking) | executive |
| Wongcharoen et al | 2017 | Counting backwards | speed |
| Wongcharoen et al | 2017 | Step width | balance |
| Wongcharoen et al | 2017 | Stride length | balance |
| Wongcharoen et al | 2017 | Verbal fluency test | language |
| Wongcharoen et al | 2017 | XcoM-BoS - Narrow walk distance | balance |
| Wongcharoen et al | 2017 | XcoM-BoS - Narrow walk speed | balance |
| Yokoyama et al | 2015 | Maximal step length | fitness |
| Yokoyama et al | 2015 | Modified Mini-Mental State (3MS) | global |
| Yokoyama et al | 2015 | Muscle strength - legs | strength |
| Yokoyama et al | 2015 | Muscle strength – cuadr. | strength |
| Yokoyama et al | 2015 | Single leg standing | balance |

| | | | |
|---|---|---|---|
| Yokoyama et al | 2015 | TMT A | speed |
| Yokoyama et al | 2015 | TUG | balance |
| You et al | 2009 | Gait stability - AP COP | balance |
| You et al | 2009 | Gait stability - ML COP | balance |
| You et al | 2009 | Gait velocity | fitness |
| You et al | 2009 | Memory recall | memory |
| Yu et al (dual-cognitive) | 2021 | Frontal Assessment Battery | global |
| Yu et al (dual-cognitive) | 2021 | Hong Kong List Learning Test-Delay Recall Trial | memory |
| Yu et al (dual-cognitive) | 2021 | Hong Kong List Learning Test-Total Learning | memory |
| Yu et al (dual-cognitive) | 2021 | Rapid Cognitive Screen | global |
| Yu et al (multicognitive) | 2021 | Frontal Assessment Battery | global |
| Yu et al (multicognitive) | 2021 | Hong Kong List Learning Test-Delay Recall Trial | memory |
| Yu et al (multicognitive) | 2021 | Hong Kong List Learning Test-Total Learning | memory |
| Yu et al (multicognitive) | 2021 | Rapid Cognitive Screen | global |

**Table S3**. Quality assessment of the reviewed articles using the Checklist for Assessing the Quality of Quantitative Studies (Kmet, Lee, and Cook (2004).

| Criteria | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Quality score | Groups |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Fabre et al. (2002) | Y | Y | Y | Y | Y | N | N | Y | N | Y | Y | Y | Y | Y | 22 | Combined; single physical; single cognitive; active control |
| Oswald et al. (2006) | Y | Y | Y | Y | N | N | N | Y | P | Y | Y | P | Y | Y | 20 | Combined; single physical, single cognitive, passive control; psychoeducation (not incl.); psychoeducation + physical (not incl.) |
| Marmeleira et al. (2009) | Y | P | Y | Y | P | N | N | Y | N | Y | Y | P | Y | Y | 19 | Combined; passive control |
| You et al. (2009) | Y | Y | Y | Y | Y | N | N | Y | N | Y | Y | P | Y | Y | 21 | Combined; physical + cognitive control |
| Legault et al. (2011) | Y | Y | Y | Y | Y | N | Y | Y | N | Y | Y | P | Y | Y | 23 | Combined; single physical; single cognitive; active control |
| Anderson-Hanley et al. (2012). | Y | Y | Y | Y | Y | N | N | Y | Y | Y | Y | P | Y | Y | 23 | Combined; single physical |
| Hiyamizu et al. (2012). | Y | Y | Y | Y | Y | Y | N | Y | N | Y | Y | P | Y | Y | 23 | Combined; single physical |
| Maillot et al. (2012) | Y | P | Y | Y | P | N | N | Y | N | Y | Y | Y | Y | Y | 20 | Combined; passive control |
| Pieramico et al. (2012) | Y | y | Y | Y | P | N | N | Y | N | Y | Y | P | Y | Y | 20 | Combined; passive control |
| Schoene et al. (2013) | Y | Y | y | y | P | Y | N | Y | P | Y | Y | Y | Y | Y | 24 | Combined, passive control |
| Shatil (2013) | Y | y | y | Y | Y | N | N | Y | P | Y | Y | P | Y | Y | 22 | Combined; single physical, single cognitive, passive control |
| Teixeira et al., (2013) | Y | Y | Y | Y | N | N | N | Y | N | Y | Y | N | Y | Y | 18 | Combined; passive control |
| Theill et al. (2013) | Y | Y | N | Y | N | N | N | Y | N | Y | Y | P | Y | Y | 17 | Combined, single cognitive, passive control |
| Linde & Alfermann (2014) | Y | Y | Y | Y | Y | Y | N | Y | N | Y | Y | P | Y | Y | 23 | Combined, single physical, single cognitive, passive control |
| McDaniel et al. (2014) | Y | Y | Y | Y | P | N | N | Y | N | Y | Y | Y | Y | Y | 21 | Combined, physical + cognitive control, cognitive + physical control, physical control + cognitive control |
| Salazar et al. (2014) | Y | Y | Y | Y | N | N | N | Y | Y | Y | Y | N | Y | Y | 20 | Combined, passive control |
| Shah et al. (2014) | Y | Y | Y | Y | N | N | N | Y | P | Y | Y | P | y | Y | 20 | Combined, single physical, single cognitive, passive control |
| Van Het Reve & de Bruin (2014) | Y | Y | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | 24 | Combined, single physical |
| Bamidis et al. (2015) | Y | Y | Y | Y | N | N | N | Y | Y | Y | Y | Y | Y | Y | 22 | Combined, passive control |
| Eggenberger et al. (2015) | Y | N | Y | Y | Y | N | Y | Y | Y | Y | Y | P | Y | Y | 23 | Combined, single physical |
| Gschwind et al. (2015) | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | 28 | Combined, passive control |
| Kitazawa et al. (2015) | Y | Y | Y | P | Y | N | N | Y | N | Y | Y | P | Y | Y | 18 | Combined, passive control |
| Ngandu et al. (2015) | Y | y | y | y | Y | y | Y | y | Y | Y | y | y | y | Y | 28 | Combined, active control |
| Nishiguchi et al (2015) | Y | Y | Y | Y | Y | Y | N | Y | Y | Y | Y | Y | Y | Y | 26 | Combined, passive control |
| Rahe et al. (2015a) | Y | P | Y | Y | N | N | N | Y | N | Y | Y | N | Y | Y | 17 | Combined, single cognitive |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Rahe et al. (2015b) | Y | P | Y | Y | N | N | N | Y | N | Y | Y | N | Y | Y | 17 | Combined, single cognitive |
| Schoene et al. (2015) | Y | Y | y | y | Y | Y | N | Y | Y | Y | Y | Y | y | Y | 26 | Combined, passive control |
| Yokoyama et al. (2015) | Y | P | Y | Y | Y | N | Y | Y | N | Y | Y | P | Y | Y | 22 | Combined, single physical |
| Desjardins-Crépeau et al. (2016) | Y | Y | Y | Y | Y | Y | N | Y | N | Y | Y | Y | Y | Y | 24 | Combined, single physical + cognitive control, single cognitive + physical control, physical control + cognitive control |
| Gill et al. (2016) | Y | N | Y | Y | Y | Y | N | Y | Y | Y | Y | P | Y | Y | 23 | Combined, single physical |
| Andrieu et al. (2017) | Y | Y | Y | Y | Y | N | P | Y | Y | Y | Y | Y | Y | Y | 25 | Combined, passive control |
| Barban et al. (2017) | Y | Y | Y | Y | Y | Y | P | Y | Y | Y | Y | Y | Y | Y | 27 | Combined, single physical, single cognitive, control |
| Jehu et al. (2017) | Y | Y | Y | Y | P | N | N | Y | N | Y | Y | P | Y | Y | 20 | Combined, single physical, passive control |
| Ng et al. (2018) | Y | P | Y | Y | Y | Y | N | Y | Y | Y | Y | N | Y | Y | 23 | Combined, single physical, single cognitive, active control |
| Wollesen et al. (2017) | Y | Y | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | 23 | Combined with and without concern of falling, passive control |
| Wongcharoen et al. (2017) | Y | Y | Y | Y | Y | N | N | Y | N | Y | Y | P | Y | Y | 20 | Combined, single physical, single cognitive |
| Htut et al. (2018) | Y | Y | Y | Y | Y | Y | N | P | Y | Y | Y | P | Y | Y | 24 | Combined, passive control |
| Laatar et al. (2018) | Y | P | Y | Y | Y | N | N | Y | N | Y | Y | P | Y | Y | 2O | Combined, single physical |
| Morita et al. (2018) | Y | P | Y | P | N | N | N | Y | N | Y | Y | N | Y | Y | 16 | Combined, passive control |
| Romera-Liebana et al. (2018) | Y | Y | Y | Y | P | Y | N | Y | Y | Y | Y | P | Y | Y | 24 | Combined, passive control |
| Joubert & Chainay, (2019) | Y | Y | Y | Y | P | N | N | Y | N | Y | Y | P | Y | Y | 20 | Combined, single cognitive, passive control |
| Norouzi et al. (2019) | Y | Y | Y | Y | P | N | N | Y | N | Y | Y | P | Y | Y | 20 | Combined, passive control, single physical (not incl.: nonequivalent exercise intervention) |
| Adcock et al. (2020) | Y | Y | Y | Y | Y | N | N | Y | N | Y | Y | N | Y | Y | 20 | Combined, passive control |
| Nilsson et al. (2020) | Y | Y | Y | Y | Y | N | Y | Y | Y | Y | Y | Y | Y | Y | 26 | Combined, single physical, single cognitive |
| Nocera et al. (2020) | Y | Y | Y | Y | P | N | N | Y | N | Y | Y | N | Y | Y | 19 | Combined, single physical, single cognitive + physical control |
| Phirom et al. (2020) | Y | Y | Y | Y | N | N | N | Y | N | Y | Y | Y | Y | Y | 20 | Combined, passive control |
| Raichlen et al. (2020) | Y | Y | Y | Y | Y | N | N | N | N | Y | Y | Y | Y | Y | 20 | Combined, single physical, single cognitive, active control |
| Takeuchi et al. (2020) | Y | Y | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | 22 | Combined, single physical, single cognitive |
| Jardim et al. (2021) | Y | Y | Y | Y | Y | Y | N | Y | Y | Y | Y | P | Y | Y | 25 | Combined, passive control |
| Yu et al. (2021) | Y | Y | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | 22 | Combined, single physical + cognitive control |

Note. Y: Yes; P: Partial; N: No; N/A: Not applicable. 1: Question / objective sufficiently described?; 2: Study design evident and appropriate?; 3: Method of subject/comparison group selection or source of information/input variables described and appropriate?; 4: Subject (and comparison group, if applicable) characteristics sufficiently described?; 5: If interventional and random allocation was possible, was it described?; 6: If interventional and blinding of investigators was possible, was it reported?; 7: If interventional and blinding of subjects was possible, was it reported?; 8: Outcome and (if applicable) exposure measure(s) well defined and robust to measurement / misclassification bias? Means of assessment reported?; 9: Sample size appropriate?; 10: Analytic methods described/justified and appropriate?; 11: Some estimate of variance is reported for the main results?; 12: Controlled for confounding; 13: Results reported in sufficient detail?; 14: Conclusions supported by the results? Yes = 2 points; Partial = 1 point; No = 0 points; N/A = not applicable

**Table S4.** Results of the influential case analysis

| Authors | ES | v | Function | Comparison | Rstudent | dffits | cook.d | cov.r | tau2.del | QE.del | hat | weight | dfbs inf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Pre-post outcomes** | | | | | | | | | | | | | |
| Oswald et al, 2016 | 1.0096 | .057 | composite | comb-control | 2.1472 | 0.1667 | 0.0272 | 0.9824 | 0.0448 | 1204.5378 | 0.0025 | 0.2481 | 0.1666 |
| Anderson-Hanley et al, 2012 | -1.5200 | .084 | memory | comb-physical | 3.5554 | 0.2944 | 0.0838 | 0.9593 | 0.0311 | 1264.8094 | 0.0017 | 0.1698 | 0.2957 |
| Anderson-Hanley et al, 2012 | 1.2280 | .077 | executive | comb-physical | 2.7594 | 0.2100 | 0.0431 | 0.9749 | 0.0326 | 1271.4589 | 0.0018 | 0.1804 | 0.2105 |
| Maillot & Hartley, 2012 | 2.0486 | .218 | speed | comb-control | 3.4601 | 0.1652 | 0.0271 | 0.9808 | 0.0332 | 1268.7936 | 0.0008 | 0.0800 | 0.1663 |
| Maillot & Hartley, 2012 | 2.1280 | .224 | executive | comb-control | 3.5739 | 0.1700 | 0.0287 | 0.9800 | 0.0332 | 1267.9221 | 0.0008 | 0.0780 | 0.1711 |
| Schoene et al (a), 2013 | 3.4110 | .338 | executive | comb-control | 5.0884 | 0.2171 | 0.0468 | 0.9712 | 0.0324 | 1254.1454 | 0.0005 | 0.0542 | 0.2196 |
| Nishiguchi et al, 2015 | -8.0860 | .834 | speed | comb-control | 8.3542 | 0.2412 | 0.0580 | 0.9662 | 0.0319 | 1210.0463 | 0.0002 | 0.0233 | 0.2450 |
| Nishiguchi et al, 2015 | -7.6470 | .754 | speed | comb-control | 8.2711 | 0.2540 | 0.0642 | 0.9639 | 0.0317 | 1211.1985 | 0.0003 | 0.0256 | 0.2582 |
| Schoene et al (b), 2015 | 9.9500 | .696 | executive | comb-control | 11.3233 | 0.4170 | 0.1724 | 0.9324 | 0.0288 | 1148.7103 | 0.0003 | 0.0277 | 0.4301 |
| Yokoyama et al, 2015 | 2.9840 | .380 | global | comb-physical | 4.1520 | 0.1524 | 0.0231 | 0.9824 | 0.0334 | 1263.9084 | 0.0005 | 0.0487 | 0.1535 |
| Wollesen et al (b), 2017 | 1.6040 | .161 | executive | comb-control | 2.9265 | 0.1553 | 0.0239 | 0.9830 | 0.0334 | 1272.2366 | 0.0010 | 0.1034 | 0.1560 |
| Norouzi et al, 2019 | 5.5560 | .534 | executive | comb-control | 6.9692 | 0.2514 | 0.0628 | 0.9646 | 0.0318 | 1231.0176 | 0.0004 | 0.0355 | 0.2553 |
| Nocera et al, 2010 | -2.9600 | .376 | executive | comb-cognitivo | 4.1320 | 0.1523 | 0.0231 | 0.9824 | 0.0334 | 1264.0738 | 0.0005 | 0.0492 | 0.1534 |
| Jardim et al, 2021 | 1.4150 | .062 | memory | comb-control | 3.6317 | 0.3573 | 0.1211 | 0.9485 | 0.0301 | 1262.3100 | 0.0021 | 0.2082 | 0.3577 |
| Jardim et al, 2021 | 2.0780 | .089 | memory | comb-control | 5.1789 | 0.5077 | 0.2418 | 0.9188 | 0.0274 | 1246.7427 | 0.0016 | 0.1630 | 0.5134 |
| Jardim et al, 2021 | 2.5250 | .104 | memory | comb-control | 6.1438 | 0.5996 | 0.3355 | 0.9006 | 0.0258 | 1234.6876 | 0.0015 | 0.1452 | 0.6108 |
| Jardim et al, 2021 | 2.6080 | .107 | memory | comb-control | 6.3031 | 0.6119 | 0.3493 | 0.8982 | 0.0256 | 1232.5822 | 0.0014 | 0.1419 | 0.6241 |
| Jardim et al, 2021 | 2.7550 | .113 | memory | comb-control | 6.5727 | 0.6306 | 0.3712 | 0.8943 | 0.0252 | 1228.9394 | 0.0014 | 0.1364 | 0.6447 |
| Jardim et al, 2021 | 3.1890 | .132 | memory | comb-control | 7.2711 | 0.6641 | 0.4134 | 0.8872 | 0.0246 | 1219.0715 | 0.0012 | 0.1208 | 0.6830 |
| Jardim et al, 2021 | 3.4670 | .146 | balance | comb-control | 5.5576 | 0.6506 | 0.3528 | 0.8106 | 0.1557 | 741.3390 | 0.0044 | 0.4406 | 0.6612 |
| Jardim et al, 2021 | 4.5690 | .211 | fitness | comb-control | 7.0074 | 0.7842 | 0.5068 | 0.7609 | 0.1408 | 724.4046 | 0.0037 | 0.3731 | 0.8176 |
| Jardim et al, 2021 | 4.9500 | .238 | fitness | comb-control | 7.3955 | 0.8000 | 0.5312 | 0.7537 | 0.1386 | 720.0702 | 0.0035 | 0.3511 | 0.8408 |
| Jardim et al, 2021 | 5.3960 | .272 | strength | comb-control | 7.7886 | 0.8047 | 0.5438 | 0.7500 | 0.1376 | 715.7461 | 0.0033 | 0.3266 | 0.8527 |
| Nishiguchi et al, 2015 | 2.015 | .131 | fitness | comb-control | 2.6774 | 0.2435 | 0.0566 | 0.9551 | 0.2228 | 738.3822 | 0.0049 | 0.4949 | 0.2440 |
| Andrieu et al, 2017 | 2.3437 | 6.67 | executive | comb-control | identified based on visual inspection of the funnel plot | | | | | | | | |
| **Pre-follow up outcomes** | | | | | | | | | | | | | |
| Norouzi et al, 2019 | 2,8620 | ,216 | executive | comb-control | identified based on visual inspection of the funnel plot | | | | | | | | |

**Table S5.** Results of the continuous and categorical moderator analyses by cognitive functions.

| | Mean difference in ES [95% CI] by moderators in cognitive outcomes | | | | | | |
|---|---|---|---|---|---|---|---|
| | **Executive functions** | **Memory** | **Attention** | **Language** | **Speed** | **Global** | **Composite** |
| **Continuous moderators** | | | | | | | |
| Quality | -0.014 [-0.042, 0.013] | -0.022 [-0.014, 0.058] | -0.028 [-0.067, 0.012] | -0.006 [-0.057, 0.056] | -0.014 [-0.068, 0.04] | -0.029 [-0.142, 0.084] | -0.075 [-0.211, 0.061] |
| Year | -0.013 [-0.037, 0.011] | -0.009 [-0.033, 0.016] | 0.003 [-0.044, 0.049] | 0.012 [-0.037, 0.058] | -0.044 [-0.094, 0.007] | 0.039 [-0.041, 0.119] | -0.053 [-0.108, -0.001] |
| N | -0.006 [-0.001, 0.000] | -0.000 [-0.001, 0.000] | -0.003 [-0.004, -0.001] *** | -0.000 [-0.001, 0.001] | -0.000 [-0.001, 0.001] | -0.001 [-0.002, 0.000] | -0.000 [-0.002, 0.001] |
| Age mean | -0.003 [-0.022, 0.016] | -0.001 [-0.022, 0.02] | -0.014 [-0.038, 0.01] | 0.015 [-0.018, 0.047] | 0.009 [-0.026, 0.044] | -0.025 [-0.089, 0.038] | 0.043 [0.025, 0.011] |
| Age SD | 0.013 [-0.034, 0.061] | 0.043 [-0.009, 0.095] | -0.019 [-0.103, 0.064] | 0.02 [-0.04, 0.08] | -0.04 [-0.123, 0.042] | -0.112 [-0.437, 0.214] | -0.001 [-0.263, 0.266] |
| Nº sessions | -0.000 [-0.003, 0.002] | -0.000 [-0.002, 0.000] | 0.001 [-0.004, 0.006] | 0.000 [-0.000, 0.001] | 0.000 [-0.001, 0.002] | -0.001 [-0.002, 0.000] | -0.001 [-0.002, 0.001] |
| Training/wks | -0.002 [-0.014, 0.011] | -0.003 [-0.01, 0.003] | -0.004 [-0.03, 0.022] | 0.001 [-0.003, 0.006] | 0.003 [-0.003, 0.009] | -0.002 [-0.009, 0.005] | -0.003 [-0.013, 0.007] |
| Minutes/week | -0.001 [-0.002, 0.000] | -0.000 [-0.001, 0.001] | -0.000 [-0.002, 0.001] | -0.001 [-0.002, 0.000] | -0.001 [-0.002, 0.000] | -0.001 [-0.005, 0.001] | -0.002 [-0.006, 0.001] |
| Min. cogn./week | -0.000 [-0.002, 0.001] | 0.000 [-0.001, 0.002] | 0.002 [-0.000, 0.005] | -0.001 [-0.003, 0.000] | -0.001 [-0.003, 0.001] | -0.001 [-0.006, 0.004] | -0.002 [-0.008, 0.007] |
| Min. phys/week | 0.000 [-0.002, 0.002] | -0.000 [-0.002, 0.02] | -0.000 [-0.001, 0.003] | -0.001 [-0.003, 0.001] | -0.001 [-0.004, 0.002] | -0.002[-0.012, 0.008] | -0.004 [-0.008, 0.000] |
| **Combinatory mode** | | | | | | | |
| Simultaneous | 0.208 [0.098, 0.318] *** | 0.154 [-0.012, 0.321] | 0.144 [0.017, 0.271] * | 0.06 [-0.18, 0.302] | 0.293 [0.1, 0.486] ** | 0.56 [-0.124, 0.996] * | NA |
| Sequential | 0.157 [-0.034, 0.348] | 0.074 [-0.088, 0.237] | 0.286 [0.071, 0.5] * | 0.023 [-0.229, 0.275] | -0.007 [-0.276, 0.262] | 0.156 [-0.592, 0.904] | 0.373 [-0.001, 0.748] |
| Separate days | 0.175 [0.001, 0.349] * | 0.16 [-0.003, 0.324] | -0.139 [-0.344, 0.067] | 0.176 [-0.037, 0.388] | 0.138[-0.242, 0.519] | 0.161 [-0.549, 0.872] | 0.003 [-0.354, 0.359] |
| **Aerobic vs non-aerobic** | | | | | | | |
| Aerobic | 0.2 [0.087, 0.313] *** | 0.108 [-0.016, 0.233] | 0.279 [0.097, 0.461] ** | 0.088 [-0.173, 0.348] | 0.175 [-0.058, 0.407] | -0.049 [-0.627, 0.53] | 0.55 [-0.179, 1.279] |
| Non-aerobic | 0.138 [0.053, 0.223] ** | 0.08 [-0.066, 0.226] | 0.032 [-0.094, 0.157] | 0.078 [-0.094, 0.25] | 0.202 [0.034, 0.37] * | 0.508 [0.149, 0.868] ** | 0.113 [-0.24, 0.465] |
| **Cognitive training type** | | | | | | | |
| Interactive[a] | 0.322 [0.179, 0.465] *** | 0.258 [-0.005, 0.521] | 0.158 [-0.095, 0.411] | NA | 0.494 [0.257, 0.731] *** | 0.56 [0.124, 0.996] * | NA |
| Computer | 0.131 [0.025, 0.227] * | 0.059 [-0.07, 0.19] | 0.069 [-0.139, 0.277] | 0.047 [-0.177, 0.271] | 0.042 [-0.152, 0.235] | 0.046 [-0.899, 0.992] | NA |
| Multicomponent[b] | 0.137 [-0.037, 0.31] | 0.196 [0.033, 0.358] * | 0.14 [-0.081, 0.362] | 0.228 [0.036, 0.421] * | 0.312 [0.009, 0.614] * | 0.206 [-0.409, 0.821] | 0.111 [-0.218, 0.447] |
| **Setting** | | | | | | | |
| Group | 0.162 [0.068, 0.256] *** | 0.182 [0.058, 0.305] ** | 0.189 [0.047, 0.331] * | 0.207 [0.012, 0.402] * | 0.241 [0.038, 0.443] * | 0.482 [0.027, 0.938] * | 0.28 [-1.033, 1.593] |
| Individual | 0.151 [0.022, 0.279] * | 0.111 [-0.117, 0.339] | 0.032 [-0.181, 0.245] | 0.086 [-0.399, 0.227] | 0.08 [-0.18, 0.34] | 0.42 [-0.378, 1.219] | NA |
| Mixed | 0.195 [-0.119, 0.51] | 0.198 [-0.023, 0.42] | NA | 0.333 [0.028, 0.638] * | 0.348 [0.015, 0.68] * | 0.162 [-0.592, 0.016] | NA |

Note. ES = Hedges' g; CI = confidence interval; NA = 3 or less effect sizes in this condition.

[a] Interactive training refers to cognitive activities that require a body-mind interaction such as exergames, square stepping, etc.

[b] Multicomponent training refers to a mixture of cognitive tasks and games, delivered in different modalities such as paper-pencil, verbally, computer-assisted, etc.

* p < .05; ** p < .01; *** p < .001.

**Table S6.** Results of the continuous and categorical moderator analyses by physical functions.

| | Mean difference in ES [95% CI] by moderators in physical outcomes | | |
|---|---|---|---|
| | **Fitness** | **Balance** | **Strength** |
| **Continuous moderators** | | | |
| Quality | -0.008 [-0.057, 0.041] | -0.039 [-0.07, -0.008] * | 0.015 [-0.066, 0.096] |
| Year | -0.035 [-0.068, -0.002] * | 0.014 [-0.025, 0.053] | -0.051 [-0.155, 0.054] |
| N | -0.000 [-0.001, 0.000] | -0.001 [-0.002, 0.000] | -0.001 [-0.004, 0.002] |
| Age mean | -0.016 [-0.06, 0.028] | -0.03 [-0.066, 0.006] | 0.038 [-0.041, 0.118] |
| Age SD | 0.029 [-0.057, 0.116] | -0.11 [-0.218, 0.002] * | -0.061 [-0.25, 0.128] |
| Nº sessions | -0.000 [-0.001, 0.000] | -0.004 [-0.01, 0.002] | -0.012 [-0.025, 0.001] |
| Training/wks | -0.002 [-0.006, 0.002] | 0.001 [-0.006, 0.006] | -0.01 [-0.022, 0.001] |
| Minutes/week | -0.000 [-0.002, 0.001] | 0.000 [-0.001, 0.002] | -0.001 [-0.003, 0.001] |
| Min. cogn./week | -0.002 [-0.006, 0.002] | 0.000 [-0.000, 0.003] | -0.001 [-0.005, 0.002] |
| Min. phys/week | -0.001 [-0.004, 0.002] | 0.001 [-0.001, 0.003] | -0.001 [-0.005, 0.003] |
| **Combinatory mode** | | | |
| Simultaneous | 0.184 [-0.021, 0.388] | 0.259 [0.153, 0.364] *** | 0.334 [0.045, 0.622] * |
| Sequential | 0.151 [-0.095, 0.398] | 0.308 [-0.008, 0.623] | 0.177 [-0.29, 0.644] |
| Separate days | 0.255 [-0.026, 0.537] | NA | 0.147 [-0.429, 0.724] |
| **Aerobic vs non-aerobic** | | | |
| Aerobic | 0.257 [0.082, 0.433] ** | 0.182 [-0.11, 0.475] | 0.373 [-0.154, 0.9] |
| Non-aerobic | 0.059 [-0.079, 0.197] | 0.272 [0.157, 0.387] *** | 0.205 [-0.01, 0.421] |
| **Cognitive training type** | | | |
| Interactive | 0.385 [0.113, 0.656] ** | 0.301 [0.154, 0.449] *** | 0.411 [0.086, 0.735] * |
| Computer | 0.04 [-0.187, 0.268] | 0.153 [-0.043, 0.343] | 0.045 [-0.273, 0.563] |
| Multicomponent | 0.288 [0.102, 0.474] ** | 0.269 [0.075, 0.464] ** | 0.4 [-0.096, 0.895] |
| **Setting** | | | |
| Group | 0.328 [0.24, 0.453] *** | 0.255 [0.12, 0.389] ** | 0.291 [0.069, 0.512] * |
| Individual | -0.073 [-0.256, 0.15] | 0.242 [0.052, 0.432] * | 0.209 [-0.034, 0.452] |
| Mixed | -0.011 [-0.255, 0.232] | 0.394 [-0.04, 0.827] | NA |

Note. ES = Hedges' g; CI = confidence interval. NA = not available due to missing effect sizes. * $p < .05$; ** $p < .01$; *** $p < .001$.